Akshat Sharma

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Education

Rajiv Gandhi Institute of Petroleum Technology **2023 – Present**
B.Tech. in Electrical Engineering (Major: E-Vehicle Technology) | Jais, Uttar Pradesh
- Minor Degree in Business Consulting

Nirmal Happy Sr. Sec. School (RBSE) **2021 – 2022**
Senior Secondary (Class XII) — Percentage: 85.6% | Karauli, Rajasthan

Technical Skills

Programming Languages: Embedded C, Python, MATLAB
Embedded & Microcontrollers: ESP32, Raspberry Pi Zero, Arduino Uno
Communication Protocols: SPI, I2C, UART
Debugging & Instrumentation: Oscilloscope, Multimeter, Serial Monitor, Logic Analyzer
Simulation & Circuit Design: MATLAB Simulink, LTSpice
EV & Power Electronics: Power Converters, Motor Control, Battery Systems(SoC Estimation)
Other Tools: Arduino IDE, VS Code, GitHub

Experience

Summer Intern – E-Vehicle Technology **May 2026 – Present**
Tata Motors | Jamshedpur, India
- Gaining hands-on exposure to Heavy Commercial Electric Vehicle (HCEV) electrical architecture, powertrain systems, vehicle communication, and manufacturing processes.
- Analyzing plant and field vehicle failures using first-principles engineering and systematic root cause analysis to identify probable causes and understand failure mechanisms.
- Assisting in the technical evaluation of failure modes and proposing feasible corrective and preventive actions to improve vehicle reliability and manufacturing quality.

Embedded Systems Intern **May 2025 – Jul 2025**
TechsageLabs | Hybrid
- Designed and developed 3+ embedded IoT applications using ESP32, Arduino, and Embedded C/C++ for real-time sensing, monitoring, and automation.
- Developed firmware for sensor and peripheral interfacing using UART, I^2C, SPI, ADC, and GPIO, with Wi-Fi-enabled MQTT communication for cloud-based data exchange.
- Performed 30+ firmware debugging, hardware validation, and system integration tests, ensuring reliable embedded system performance and prototype stability.

Projects

FOC-Controlled 3ph VSI-Fed PMSM Drive **Mar 2026**
MATLAB Simulink, Motor Control
- Modelled 3-phase PMSM from first principles in the dq rotating frame — deriving voltage equations, flux linkage, electromagnetic torque, and mechanical dynamics without toolbox dependencies.
- Implemented Clarke/Park transforms and SVPWM (sector detection, adjacent-vector timing, zero-vector insertion) from scratch; designed cascaded PI current and speed loops with decoupling feedforward and anti-windup.
- Validated full FOC against V/f baseline through load step and speed reversal tests, confirming stable torque response and speed tracking across rated operating conditions.

Carbon-Aware Microgrid Optimization System **Jan 2026**
ESP32, Python, MQTT, LSTM/XGBoost
- Engineered ESP32-based microgrid with 3 relay-controlled sources (solar/battery/grid) achieving 63%+ simulated CO2 reduction across 50 test scenarios through ML-driven carbon prediction.
- Calibrated ACS712 current sensors and voltage dividers to achieve ±5% power measurement accuracy across 0-25V range with real-time MQTT telemetry.
- Implemented LSTM/XGBoost models achieving 90% load forecasting accuracy and embedded fail-safe protocols preventing backfeed with 100% relay transition reliability over 200+ operations.
- Earned 3rd Prize at RGIPT Institute Day (Top 3 out of 20+ projects) competing against 15+ teams.

2.4 GHz Wireless Interference Analysis using ESP32 **Nov 2025**
ESP32, nRF24L01, Arduino IDE, SPI
- Designed interference analysis system scanning 20+ channels (2400-2480MHz) with dual nRF24L01 modules achieving 95% packet transmission reliability across 500 test packets.
- Implemented channel sweeping algorithms to identify interference patterns across 2.4GHz spectrum with 1ms sampling resolution and 5dBm sensitivity.
- Analyzed cross-interference effects between Bluetooth and WiFi across 10 different configurations, documenting 5-page findings that improved wireless coexistence strategies.

Positions of Responsibility

Coordinator, Quiz Club, RGIPT **2025 – Present**
- Leading club of 30+ members, organizing 5+ inter-college events with 200+ total participants.

Teaching Volunteer, Arpan: Social Services Council, RGIPT **2024 – 2025**
- Provided academic mentoring to 15+ underprivileged students, improving their average grades by 20%.

Certifications and Achievements

- 3rd Prize, Model Prototype Display | Institute Day, RGIPT (15+ participating teams)
- Gate Driving Circuit Design For Power Converters | EEE Department, RGIPT
- Joint Faculty Development Program in EV Tech | EECT-IIT Roorkee
- Modular course on FOC-based PMSM motor-control | EEE Department, RGIPT